

6.1

a) $A^2 = A \Rightarrow \lambda \in \{0, 1\}$ *plach'*

$$Ax = \lambda x \quad x \neq 0$$

$$A^2 x = A(Ax) = A(\lambda x) = \lambda(Ax) = \lambda^2 x$$

$$\downarrow A^2 = A$$

$$Ax = \lambda^2 x$$

$$\lambda x = \lambda^2 x$$

$$\lambda^2 x - \lambda x = 0$$

$$(\lambda^2 - \lambda)x = 0, \quad x \neq 0$$

$$\lambda^2 - \lambda = \lambda(\lambda - 1) = 0$$

$$\lambda \in \{0, 1\}$$

b) *neplach'*

$$A = I_n$$

$$I_n^2 = I_n \quad \forall n \in \mathbb{N}, \text{ ale } \lambda \in \{1\}$$

c) *plach'*

$$Ax = \lambda x$$

$$A^2 x = A(Ax) = A(\lambda x) = \lambda(Ax) = \lambda^2 x$$

$$A^3 x = A(A^2 x) = A(\lambda^2 x) = \lambda^2(Ax) = \lambda^3 x$$

$$\vdots$$

$$A^n x = \lambda^n x = 0, \quad x \neq 0$$

$$\lambda^n = 0$$

$$\lambda = 0$$

d) platí $c) \Rightarrow d)$, protože každá reálná
čtvercová matice má alespoň jedno skalar. číslo

$\hookrightarrow f_A(\lambda) = \det(A - \lambda I)$ polynom st. k po A a
ke základní věty algebry: každý nemulý polynom st. ≥ 1
má alespoň jeden komplexní kořen